

OakSol Commerce

E-Commerce Platform
Database Architecture

Phase 1 — Multi-Tenant PostgreSQL Schema

32	8	19	16
Tables	Groups	Relationships	Indexes

Prepared for	Mokasafat Amir Behesti (Boss)
Prepared by	Aftab OakSol Commerce
Date	June 4, 2026
Database	PostgreSQL 16+
Architecture	Multi-Tenant · UUID PKs · Row-Level Isolation

Table of Contents

■ Shop & Tenant

shops [NEW]

shop_domains [NEW]

■ System

users [MOD]

settings

activity_logs [NEW]

■ Product Catalog

categories

brands

products

product_gallery

product_variants

variant_attributes

product_sections

product_faqs

■ Inventory

warehouses [NEW]

inventory_logs [MOD]

■ Customer

customers [MOD]

addresses

■ Cart & Order

carts

cart_items

orders

order_items

order_status_logs [NEW]

shipping_rates [NEW]

coupons

■ Payment

payment_gateways

payments

taxes [NEW]

■ Content & Marketing

reviews

notifications [NEW]

Indexes & Performance — 16 optimised indexes for multi-tenant queries

Entity-Relationship Map — 19 relationships with cardinality

Shop & Tenant

shops [NEW]

Every merchant account — core of multi-tenancy

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
name	VARCHAR(150) NOT NULL	Display name of the shop
slug	VARCHAR(100) UNIQUE NOT NULL	URL-safe identifier
owner_id	UUID REFERENCES users(id) NOT NULL	FK → users
plan	VARCHAR(50) DEFAULT 'free'	free / starter / pro
status	VARCHAR(30) DEFAULT 'active'	active / suspended / deleted
logo_url	TEXT	CDN URL
description	TEXT	Short shop bio
currency	CHAR(3) DEFAULT 'BDT'	ISO 4217
timezone	VARCHAR(60) DEFAULT 'Asia/Dhaka'	
created_at	TIMESTAMPTZ DEFAULT NOW()	
updated_at	TIMESTAMPTZ DEFAULT NOW()	

shop_domains [NEW]

Subdomain + custom domain mapping per shop

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	FK → shops
domain	VARCHAR(255) UNIQUE NOT NULL	e.g. aftabshop.oaksolcommerce.in
type	VARCHAR(20) DEFAULT 'subdomain'	subdomain / custom
is_primary	BOOLEAN DEFAULT FALSE	One primary per shop
status	VARCHAR(30) DEFAULT 'pending'	pending / active / failed
verified_at	TIMESTAMPTZ NULL	Until DNS verified
created_at	TIMESTAMPTZ DEFAULT NOW()	

System

users [MOD]

Admin & staff accounts (per shop)

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	FK → shops
name	VARCHAR(150) NOT NULL	
email	VARCHAR(255) UNIQUE NOT NULL	
password_hash	TEXT NOT NULL	
role	VARCHAR(30) DEFAULT 'staff'	admin / staff
avatar_url	TEXT	
is_active	BOOLEAN DEFAULT TRUE	
last_login_at	TIMESTAMPTZ	
created_at	TIMESTAMPTZ DEFAULT NOW()	
updated_at	TIMESTAMPTZ DEFAULT NOW()	

settings

Key-value store for shop-level config

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	FK → shops
key	VARCHAR(100) NOT NULL	e.g. 'smtp_host'
value	TEXT	JSON or plain string
group	VARCHAR(60)	email / payment / seo
updated_at	TIMESTAMPTZ DEFAULT NOW()	
—	UNIQUE(shop_id, key)	

activity_logs [NEW]
Admin action audit trail

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	FK → shops
user_id	UUID REFERENCES users(id)	FK → users
action	VARCHAR(100) NOT NULL	e.g. 'order.status_updated'
entity_type	VARCHAR(60)	order / product / user
entity_id	UUID	
metadata	JSONB DEFAULT '{}'	Extra context
ip_address	INET	
created_at	TIMESTAMPTZ DEFAULT NOW()	

Product Catalog

categories

Nested category tree (self-referential)

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	FK → shops
parent_id	UUID REFERENCES categories(id)	NULL = root
name	VARCHAR(150) NOT NULL	
slug	VARCHAR(150) NOT NULL	
image_url	TEXT	
sort_order	INT DEFAULT 0	
is_active	BOOLEAN DEFAULT TRUE	
created_at	TIMESTAMPTZ DEFAULT NOW()	
—	UNIQUE(shop_id, slug)	

brands

Product brands / manufacturers

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	FK → shops
name	VARCHAR(150) NOT NULL	
slug	VARCHAR(150) NOT NULL	
logo_url	TEXT	
is_active	BOOLEAN DEFAULT TRUE	
created_at	TIMESTAMPTZ DEFAULT NOW()	
—	UNIQUE(shop_id, slug)	

products

Core product record

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	FK → shops
category_id	UUID REFERENCES categories(id)	
brand_id	UUID REFERENCES brands(id)	
name	VARCHAR(255) NOT NULL	
slug	VARCHAR(255) NOT NULL	
description	TEXT	
short_desc	TEXT	For listing cards
price	NUMERIC(12,2) NOT NULL	Display price
compare_price	NUMERIC(12,2)	Crossed-out original
cost_price	NUMERIC(12,2)	Internal only
master_sku	VARCHAR(100)	Optional master SKU
status	VARCHAR(30) DEFAULT 'draft'	draft / active / archived
is_featured	BOOLEAN DEFAULT FALSE	
has_variants	BOOLEAN DEFAULT FALSE	
tax_id	UUID REFERENCES taxes(id)	
total_sold	INT DEFAULT 0	
meta_title	VARCHAR(255)	
meta_description	TEXT	
created_at	TIMESTAMPTZ DEFAULT NOW()	
updated_at	TIMESTAMPTZ DEFAULT NOW()	
—	UNIQUE(shop_id, slug)	

product_gallery

Product images (multiple per product)

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
product_id	UUID REFERENCES products(id) ON DELETE CASCADE	
url	TEXT NOT NULL	
alt_text	TEXT	
sort_order	INT DEFAULT 0	
is_cover	BOOLEAN DEFAULT FALSE	
created_at	TIMESTAMPTZ DEFAULT NOW()	

product_variants

Each SKU with its own price & stock

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
product_id	UUID REFERENCES products(id) ON DELETE CASCADE	
sku	VARCHAR(100) UNIQUE NOT NULL	
label	VARCHAR(255)	e.g. 'Red / XL'
price	NUMERIC(12,2) NOT NULL	
compare_price	NUMERIC(12,2)	
cost_price	NUMERIC(12,2)	
stock_qty	INT DEFAULT 0 NOT NULL	
low_stock_at	INT DEFAULT 5	Alert threshold
image_url	TEXT	
total_sold	INT DEFAULT 0	
is_active	BOOLEAN DEFAULT TRUE	
sort_order	INT DEFAULT 0	
created_at	TIMESTAMPTZ DEFAULT NOW()	
updated_at	TIMESTAMPTZ DEFAULT NOW()	

variant_attributes

Key-value attributes for each variant

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
variant_id	UUID REFERENCES product_variants(id) ON DELETE CASCADE	
attr_key	VARCHAR(60) NOT NULL	e.g. 'color', 'size'
attr_value	VARCHAR(150) NOT NULL	e.g. 'Red', 'XL'
sort_order	INT DEFAULT 0	

product_sections

Homepage & page layout sections

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
title	VARCHAR(200) NOT NULL	
type	VARCHAR(50)	banner / grid / carousel
config	JSONB DEFAULT '{}'	
sort_order	INT DEFAULT 0	
is_active	BOOLEAN DEFAULT TRUE	
created_at	TIMESTAMPTZ DEFAULT NOW()	

product_faqs

Q&A; per product

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
product_id	UUID REFERENCES products(id) ON DELETE CASCADE	
question	TEXT NOT NULL	
answer	TEXT NOT NULL	
sort_order	INT DEFAULT 0	
created_at	TIMESTAMPTZ DEFAULT NOW()	

Inventory

warehouses [NEW]

Warehouse / location reference

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
name	VARCHAR(150) NOT NULL	e.g. 'Main Warehouse'
address	TEXT	
is_active	BOOLEAN DEFAULT TRUE	
created_at	TIMESTAMPTZ DEFAULT NOW()	

inventory_logs [MOD]

Stock change audit trail per variant

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
variant_id	UUID REFERENCES product_variants(id)	
warehouse_id	UUID REFERENCES warehouses(id)	FK added — MOD
type	VARCHAR(30) NOT NULL	restock / sale / adjustment
qty_change	INT NOT NULL	+ve in, -ve out
qty_after	INT NOT NULL	Stock snapshot
ref_id	UUID	order_id / NULL
note	TEXT	
created_by	UUID REFERENCES users(id)	
created_at	TIMESTAMPTZ DEFAULT NOW()	

Customer

customers [MOD]

Registered & guest shoppers

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
name	VARCHAR(150)	
email	VARCHAR(255)	
phone	VARCHAR(30)	
password_hash	TEXT	NULL for guests
guest_token	VARCHAR(255) UNIQUE	Session token for guests
is_verified	BOOLEAN DEFAULT FALSE	
avatar_url	TEXT	
total_orders	INT DEFAULT 0	
total_spent	NUMERIC(14,2) DEFAULT 0	
created_at	TIMESTAMPTZ DEFAULT NOW()	
updated_at	TIMESTAMPTZ DEFAULT NOW()	
—	UNIQUE(shop_id, email) WHERE email IS NOT NULL	

addresses

Shipping & billing addresses

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
customer_id	UUID REFERENCES customers(id) ON DELETE CASCADE	
type	VARCHAR(20) DEFAULT 'shipping'	shipping / billing
full_name	VARCHAR(150) NOT NULL	
phone	VARCHAR(30)	
address_line1	TEXT NOT NULL	
address_line2	TEXT	
city	VARCHAR(100) NOT NULL	
state	VARCHAR(100)	
postal_code	VARCHAR(20)	
country	CHAR(2) DEFAULT 'BD'	ISO 3166-1 alpha-2
is_default	BOOLEAN DEFAULT FALSE	
created_at	TIMESTAMPTZ DEFAULT NOW()	

Cart & Order

carts

Active shopping carts (registered + guest)

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
customer_id	UUID REFERENCES customers(id)	NULL for guests
session_id	VARCHAR(255) UNIQUE	Guest cart token
coupon_id	UUID REFERENCES coupons(id)	
expires_at	TIMESTAMPTZ	
created_at	TIMESTAMPTZ DEFAULT NOW()	
updated_at	TIMESTAMPTZ DEFAULT NOW()	

cart_items

Line items inside a cart

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
cart_id	UUID REFERENCES carts(id) ON DELETE CASCADE	
variant_id	UUID REFERENCES product_variants(id)	
qty	INT NOT NULL DEFAULT 1 CHECK (qty > 0)	
unit_price	NUMERIC(12,2) NOT NULL	Price at time of add
created_at	TIMESTAMPTZ DEFAULT NOW()	
—	UNIQUE(cart_id, variant_id)	

orders

Placed orders — financial record

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
customer_id	UUID REFERENCES customers(id)	
order_number	VARCHAR(30) UNIQUE NOT NULL	Human-readable #
status	VARCHAR(30) DEFAULT 'pending'	pending / confirmed / shipped / cancelled
subtotal	NUMERIC(12,2) NOT NULL	
discount_amount	NUMERIC(12,2) DEFAULT 0	
shipping_amount	NUMERIC(12,2) DEFAULT 0	
tax_amount	NUMERIC(12,2) DEFAULT 0	
total	NUMERIC(12,2) NOT NULL	
coupon_id	UUID REFERENCES coupons(id)	
shipping_rate_id	UUID REFERENCES shipping_rates(id)	
shipping_address	JSONB NOT NULL	Snapshot at order time
billing_address	JSONB	
notes	TEXT	
created_at	TIMESTAMPTZ DEFAULT NOW()	
updated_at	TIMESTAMPTZ DEFAULT NOW()	

order_items

Product snapshot per order line

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
order_id	UUID REFERENCES orders(id) ON DELETE CASCADE	
variant_id	UUID REFERENCES product_variants(id)	
product_snap	JSONB NOT NULL	Name, SKU, image at order time
qty	INT NOT NULL	
unit_price	NUMERIC(12,2) NOT NULL	
line_total	NUMERIC(12,2) NOT NULL	
created_at	TIMESTAMPTZ DEFAULT NOW()	

order_status_logs [NEW]

Full audit trail of order status changes

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
order_id	UUID REFERENCES orders(id) ON DELETE CASCADE	
from_status	VARCHAR(30)	NULL for first entry
to_status	VARCHAR(30) NOT NULL	
note	TEXT	
changed_by	UUID REFERENCES users(id)	
created_at	TIMESTAMPTZ DEFAULT NOW()	

shipping_rates [NEW]

Flat-rate or zone-based delivery pricing

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
name	VARCHAR(150) NOT NULL	e.g. 'Inside Dhaka'
rate	NUMERIC(10,2) NOT NULL	
free_above	NUMERIC(10,2) DEFAULT 0	Free shipping threshold
zone_label	VARCHAR(100)	Geo zone label
is_active	BOOLEAN DEFAULT TRUE	
created_at	TIMESTAMPTZ DEFAULT NOW()	

coupons

Discount codes

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
code	VARCHAR(60) NOT NULL	
type	VARCHAR(20) NOT NULL	percent / fixed / free_shipping
value	NUMERIC(10,2) NOT NULL	
min_order	NUMERIC(10,2) DEFAULT 0	
usage_limit	INT	NULL = unlimited
used_count	INT DEFAULT 0	
starts_at	TIMESTAMPTZ	
ends_at	TIMESTAMPTZ	
is_active	BOOLEAN DEFAULT TRUE	
created_at	TIMESTAMPTZ DEFAULT NOW()	
—	UNIQUE(shop_id, code)	

Payment

payment_gateways

Gateway config per shop (bKash, card, COD)

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
name	VARCHAR(100) NOT NULL	bKash / SSLCommerz / COD
slug	VARCHAR(60) NOT NULL	
config	JSONB DEFAULT '{}'	API keys — encrypted at rest
is_active	BOOLEAN DEFAULT TRUE	
sort_order	INT DEFAULT 0	
created_at	TIMESTAMPTZ DEFAULT NOW()	

payments

Transaction record per order

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
order_id	UUID REFERENCES orders(id)	
gateway_id	UUID REFERENCES payment_gateways(id)	
amount	NUMERIC(12,2) NOT NULL	
currency	CHAR(3) DEFAULT 'BDT'	
status	VARCHAR(30) DEFAULT 'pending'	pending / paid / failed / refunded
transaction_id	VARCHAR(255) UNIQUE	Gateway reference
gateway_resp	JSONB DEFAULT '{}'	Raw gateway response
paid_at	TIMESTAMPTZ	
created_at	TIMESTAMPTZ DEFAULT NOW()	

taxes [NEW]

VAT / GST rate definitions

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
name	VARCHAR(100) NOT NULL	e.g. 'VAT 15%'
rate	NUMERIC(5,2) NOT NULL	15.00 = 15%
type	VARCHAR(30) DEFAULT 'percent'	
is_inclusive	BOOLEAN DEFAULT FALSE	
is_active	BOOLEAN DEFAULT TRUE	
created_at	TIMESTAMPTZ DEFAULT NOW()	

Content & Marketing

reviews

Customer product reviews & ratings

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
product_id	UUID REFERENCES products(id) ON DELETE CASCADE	
customer_id	UUID REFERENCES customers(id)	
rating	SMALLINT NOT NULL CHECK (rating BETWEEN 1 AND 5)	
title	VARCHAR(200)	
body	TEXT	
status	VARCHAR(20) DEFAULT 'pending'	pending / approved / rejected
created_at	TIMESTAMPTZ DEFAULT NOW()	

notifications [NEW]

Order & system alerts per customer

Column	Type / Constraint	Note
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()	
shop_id	UUID REFERENCES shops(id) ON DELETE CASCADE	
customer_id	UUID REFERENCES customers(id) ON DELETE CASCADE	
type	VARCHAR(60) NOT NULL	order_confirmed / shipped / ...
title	VARCHAR(255) NOT NULL	
body	TEXT	
is_read	BOOLEAN DEFAULT FALSE	
ref_id	UUID	order_id or NULL
created_at	TIMESTAMPTZ DEFAULT NOW()	

Indexes & Performance

Every query in a multi-tenant system must be prefixed by `shop_id`. The indexes below ensure PostgreSQL never performs a full table scan.

Index Name	On	Purpose
<code>idx_products_shop</code>	<code>products(shop_id)</code>	Multi-tenancy filter
<code>idx_orders_shop</code>	<code>orders(shop_id)</code>	Multi-tenancy filter
<code>idx_customers_shop</code>	<code>customers(shop_id)</code>	Multi-tenancy filter
<code>idx_variants_product</code>	<code>product_variants(product_id)</code>	
<code>idx_variants_sku</code>	<code>product_variants(sku)</code>	
<code>idx_cart_items_cart</code>	<code>cart_items(cart_id)</code>	
<code>idx_order_items_order</code>	<code>order_items(order_id)</code>	
<code>idx_payments_order</code>	<code>payments(order_id)</code>	
<code>idx_inventory_variant</code>	<code>inventory_logs(variant_id)</code>	
<code>idx_reviews_product</code>	<code>reviews(product_id)</code>	
<code>idx_notifications_cust</code>	<code>notifications(customer_id, is_read)</code>	
<code>idx_activity_shop_time</code>	<code>activity_logs(shop_id, created_at DESC)</code>	
<code>idx_shop_domain</code>	<code>shop_domains(domain) UNIQUE</code>	Every request — multi-tenant routing
<code>idx_order_number</code>	<code>orders(shop_id, order_number) UNIQUE</code>	Customer-facing tracking
<code>idx_cart_session</code>	<code>carts(session_id) WHERE session_id IS NOT NULL</code>	Guest cart lookup
<code>idx_variant_stock</code>	<code>product_variants(shop_id, stock_qty) WHERE stock_qty <= low_stock_at</code>	Low-stock alerts

Entity Relationships

From	To	Description	Cardinality
shops	users	1 shop → many staff users	1 : N
shops	shop_domains	1 shop → many domains	1 : N
shops	products	1 shop → many products	1 : N
products	product_variants	1 product → many variants (SKUs)	1 : N
product_variants	variant_attributes	1 variant → many attributes	1 : N
product_variants	inventory_logs	1 variant → full stock history	1 : N
warehouses	inventory_logs	1 warehouse → many stock events	1 : N
customers	carts	1 customer → 1 active cart	1 : 1
carts	cart_items	1 cart → many items	1 : N
customers	orders	1 customer → many orders	1 : N
orders	order_items	1 order → many line items	1 : N
orders	order_status_logs	1 order → full status history	1 : N
orders	payments	1 order → 1+ payment attempts	1 : N
payment_gateways	payments	1 gateway → many transactions	1 : N
products	reviews	1 product → many reviews	1 : N
customers	notifications	1 customer → many notifications	1 : N
taxes	products	1 tax rate → many products	1 : N
shipping_rates	orders	1 shipping rate → many orders	1 : N
coupons	orders	1 coupon → many orders	1 : N

Key Design Decisions

UUID Primary Keys	All PKs use gen_random_uuid() — safe for distributed systems, no sequential enumeration risk.
Multi-Tenant Isolation	Every table carries shop_id. Row-level isolation — no cross-tenant data leak possible at DB layer.
Custom Domain Routing	shop_domains handles both subdomains (aftabshop.oaksolcommerce.in) and custom domains with SSL status tracking.
Variant Stock System	product_variants holds per-SKU stock. inventory_logs gives a full audit trail. low_stock_at enables automated alerts.
Guest Checkout	customers.guest_token + carts.session_id enables guest checkout. Guests can be merged into registered accounts later.

Order Snapshot Pattern	order_items.product_snap (JSONB) stores product name/price/image at order time — price changes never corrupt history.
Address Snapshot	orders.shipping_address (JSONB) snapshots the address — customers can change addresses without breaking old orders.
Soft Deletes	Use status/is_active columns rather than hard deletes — preserves referential integrity and history.
TIMESTAMPTZ	All timestamps use TIMESTAMPTZ (timezone-aware) — essential for multi-region and DST correctness.